1)ARCHITECTURE DE BASE

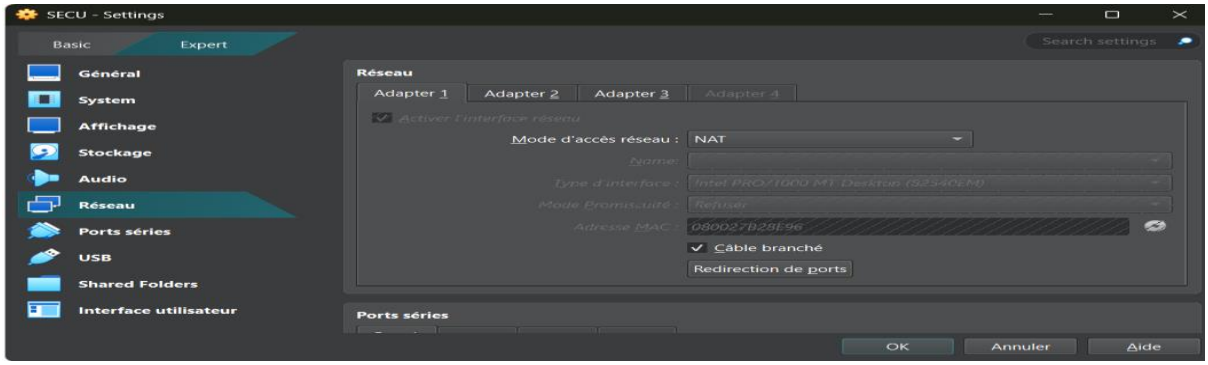
2) INSTALLATION ET CONFIGURATION DE PF-SENSE

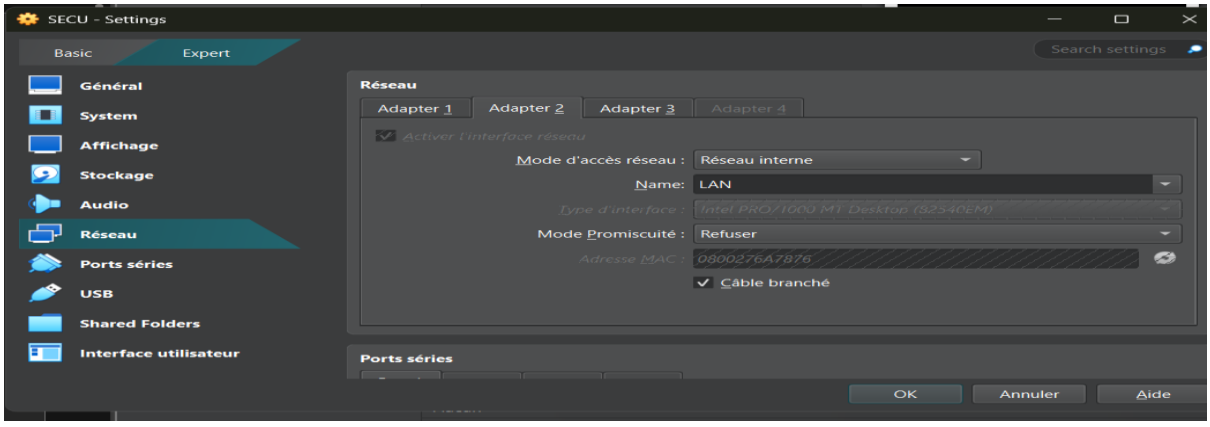
2.1. Installation de pfSense

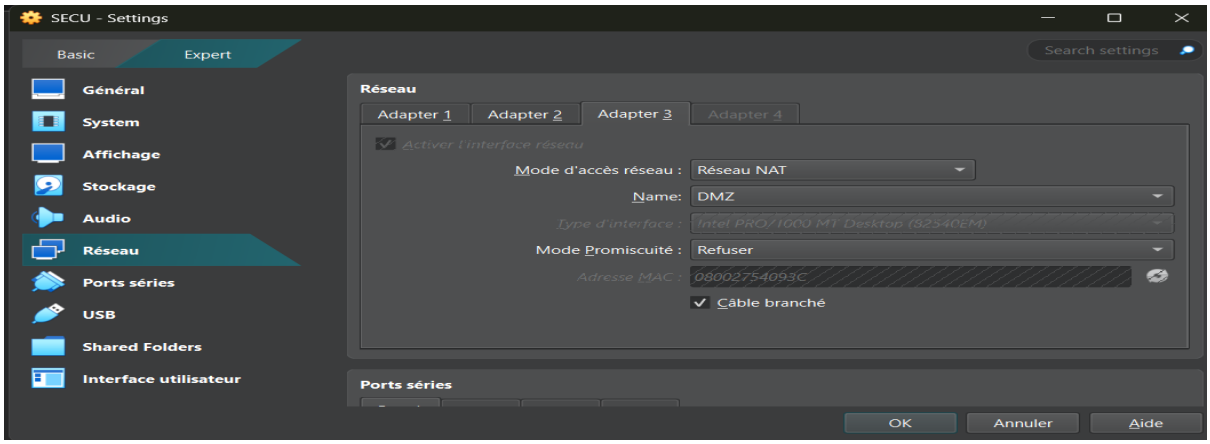
• Réseau interne (LAN) : Contient les postes de travail des utilisateurs (192.168.1.0/24)

• DMZ : Zone pour les services accessibles depuis l'extérieur (serveur web, serveur de messagerie, etc.) (192.168.2.0/24)

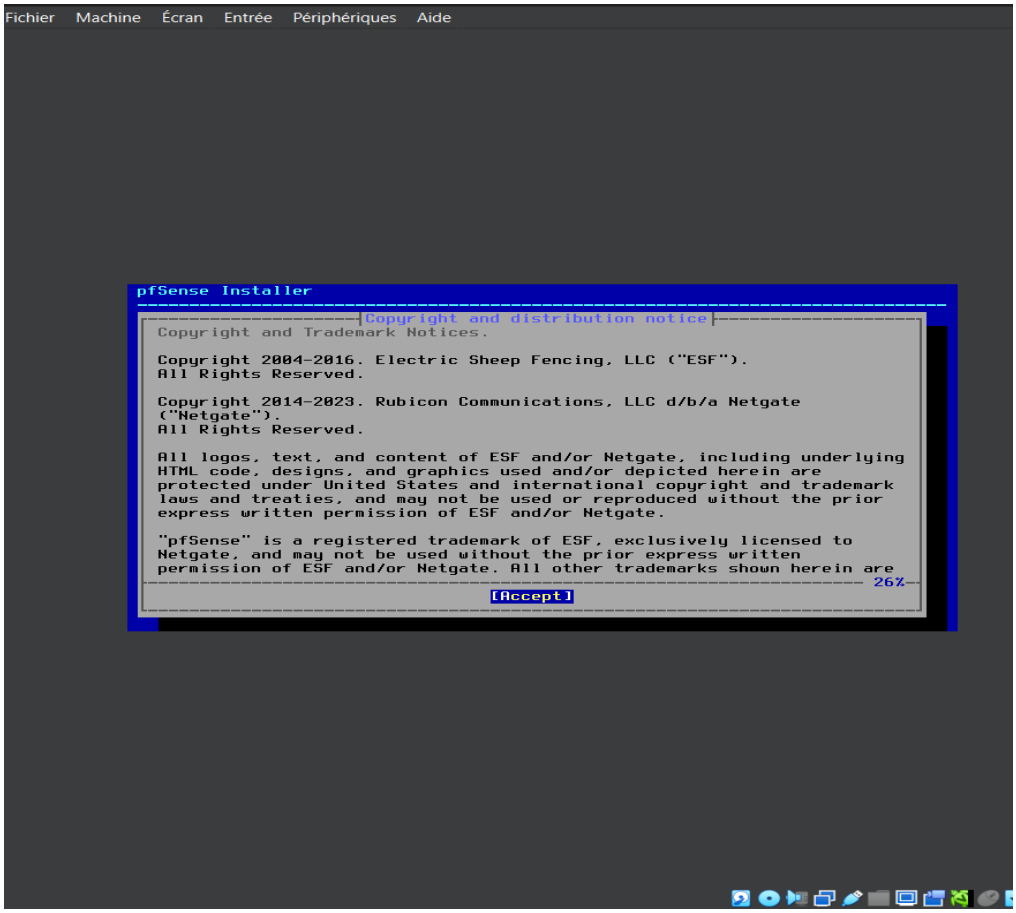
• Réseau externe (WAN) ou (NAT): Connexion à Internet, fournie par un modem ou un routeur.

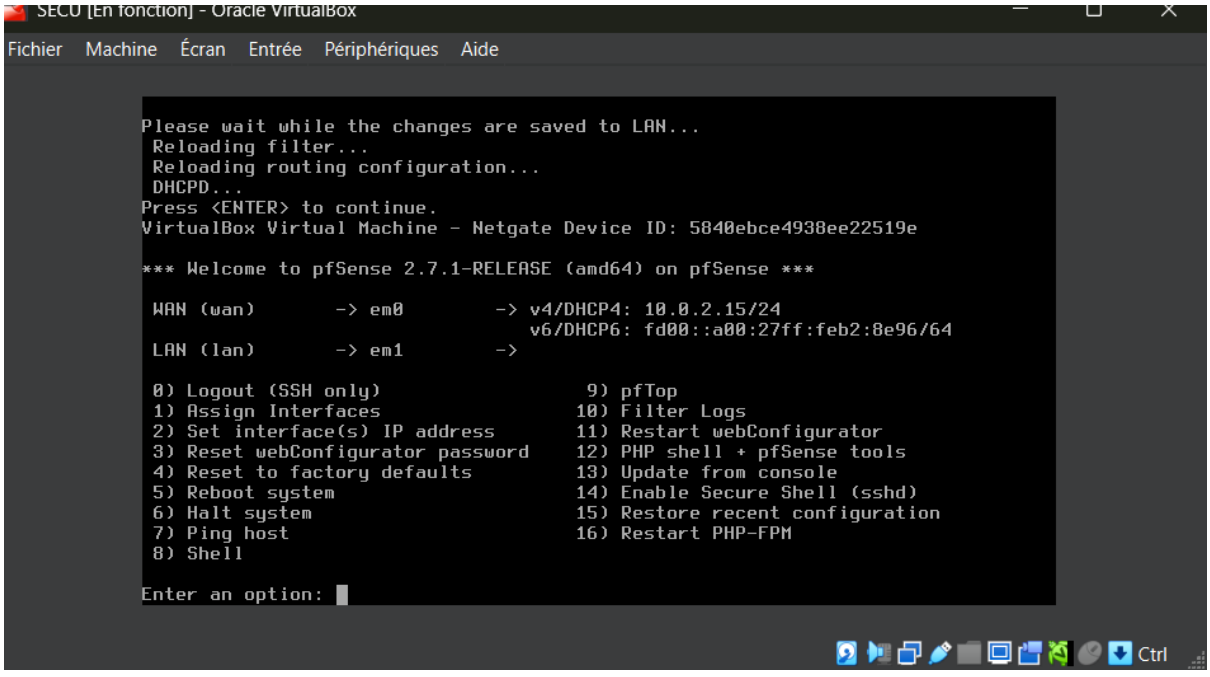




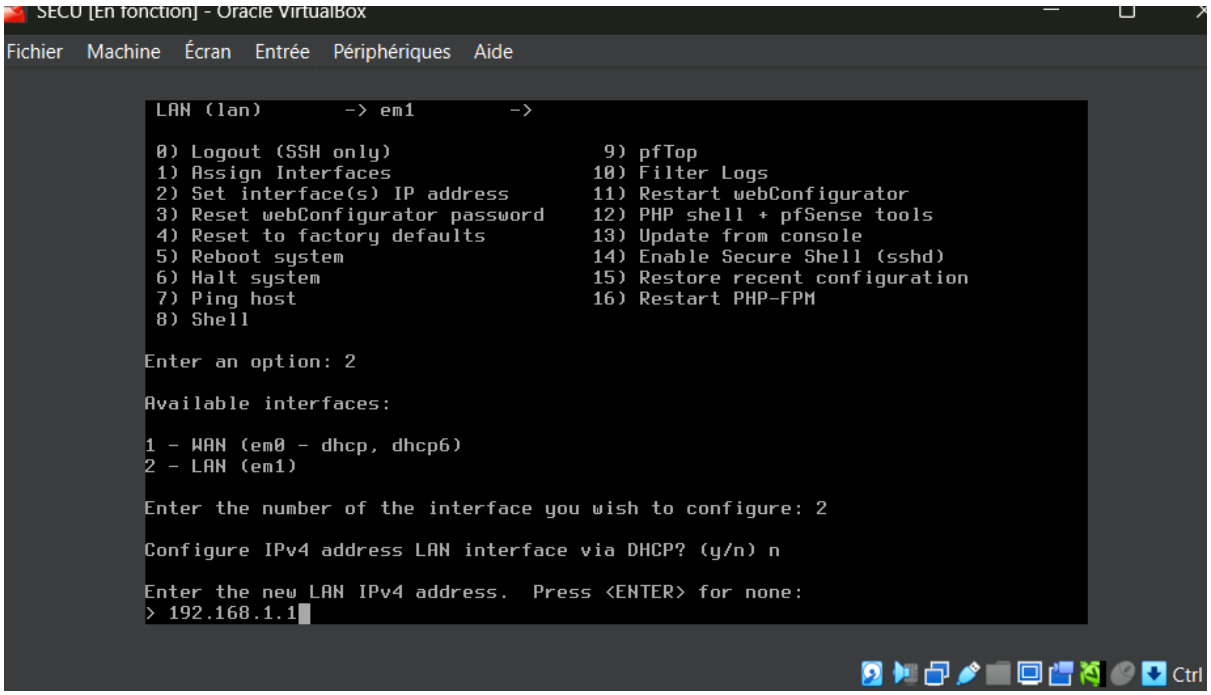


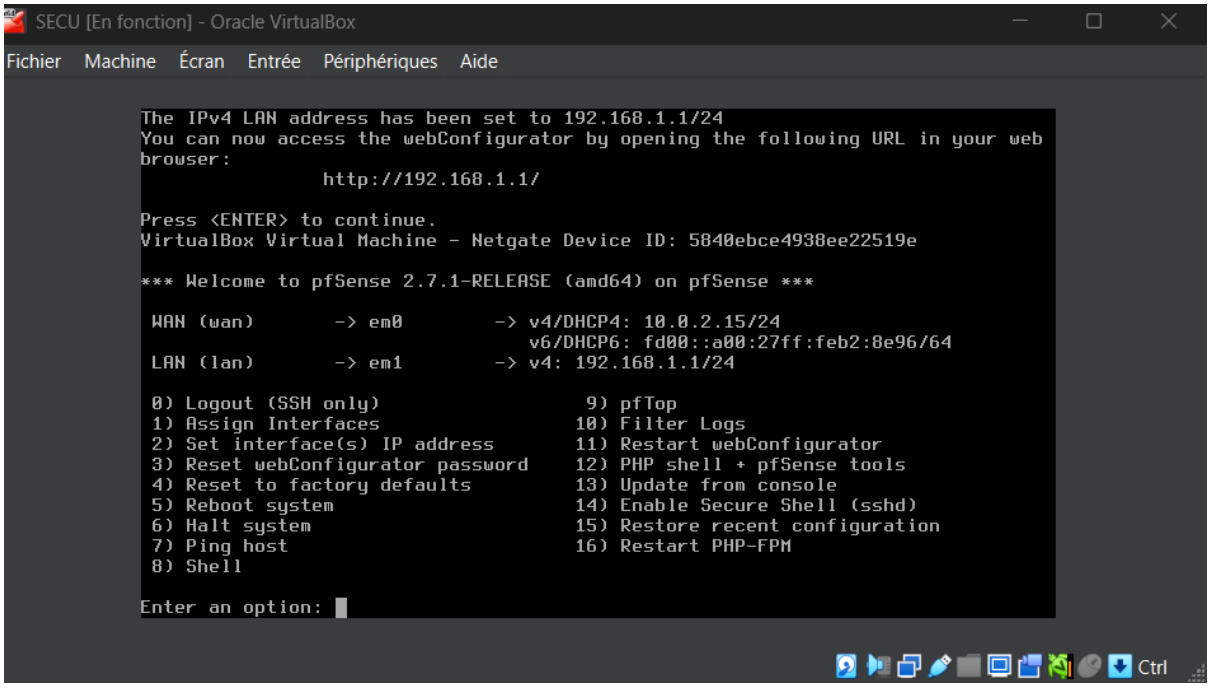
2.2. Configuration initiale de pfSense



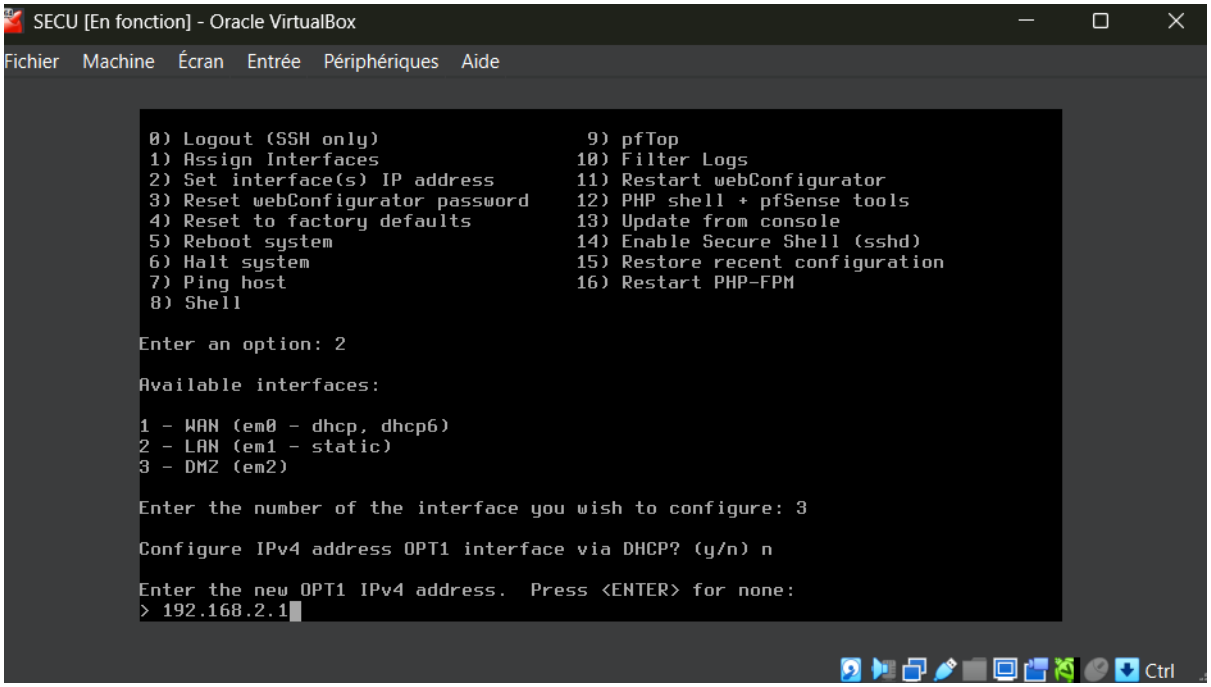


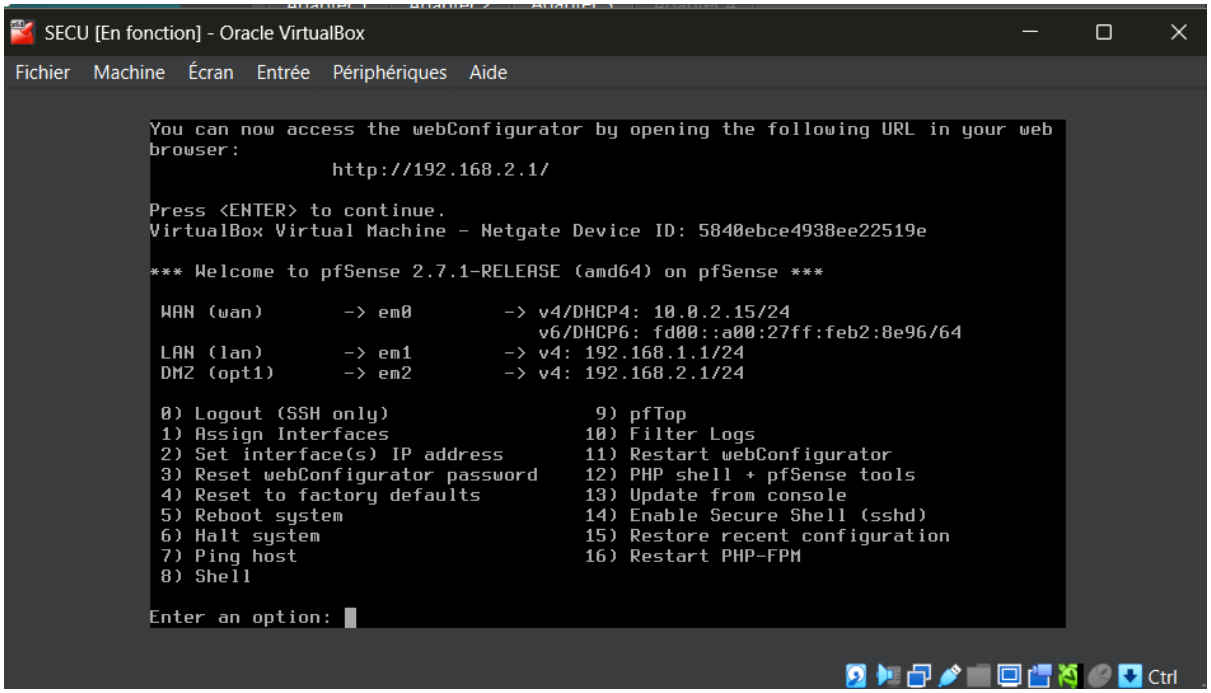
Une fois linterface acquise nous allons donner a la machine linterface demander dans le tp



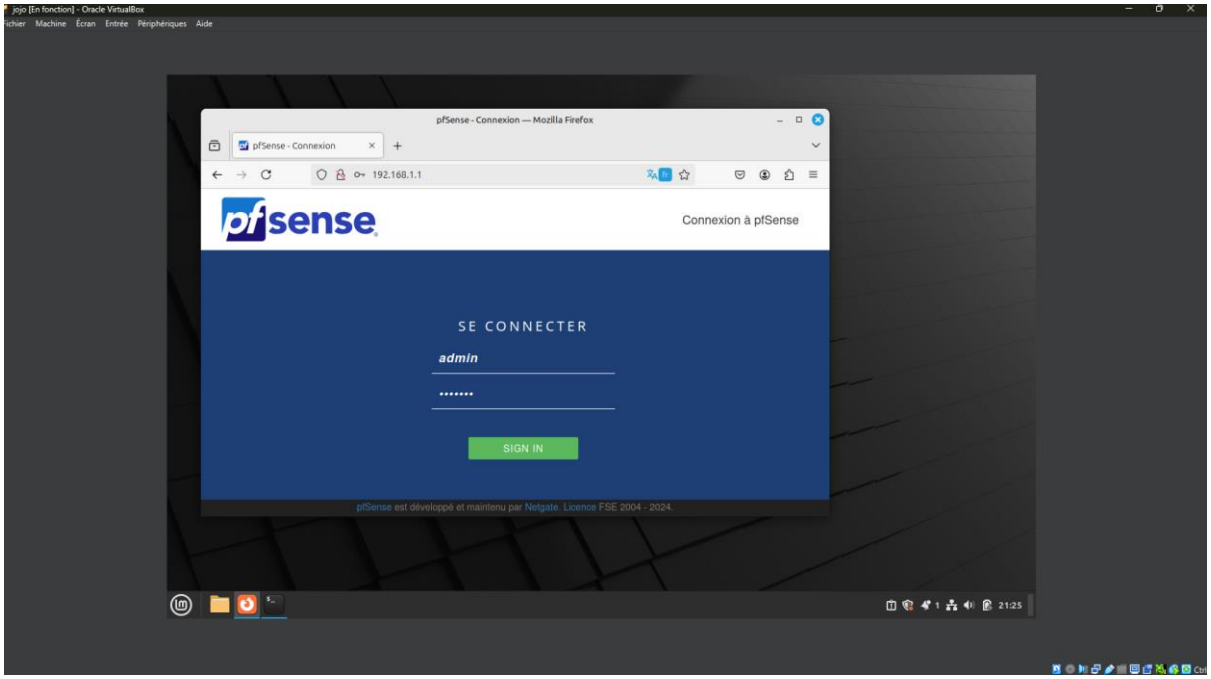


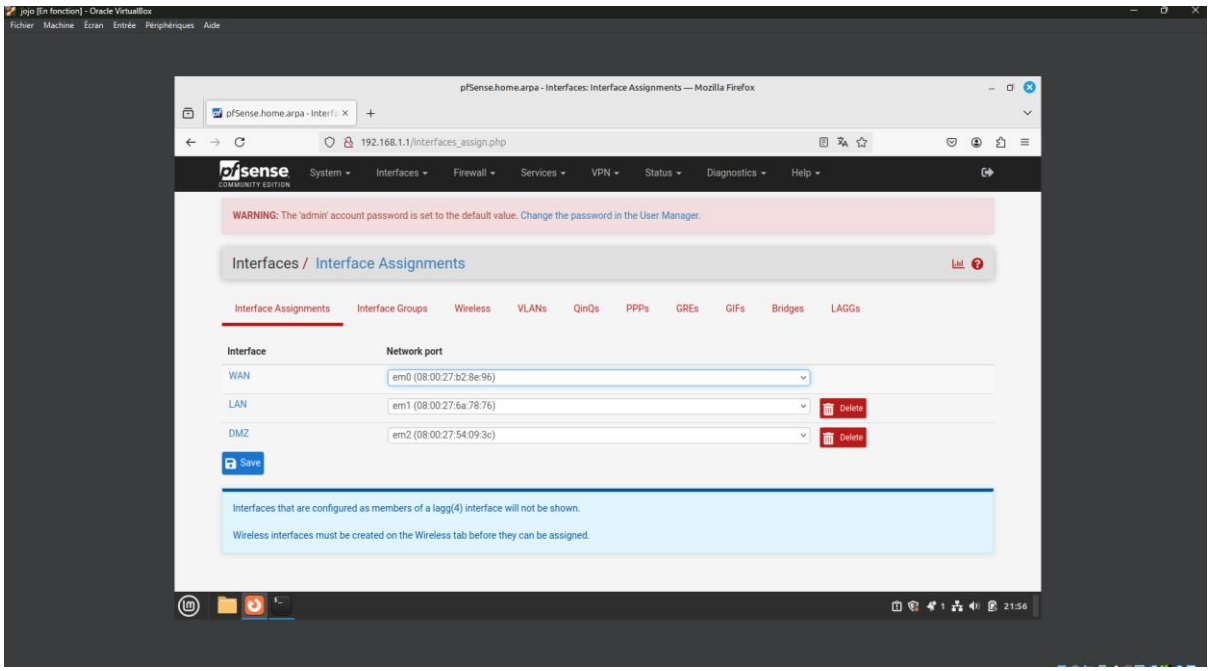
Ensuiste nous allons assigner une addresse pour la dmz



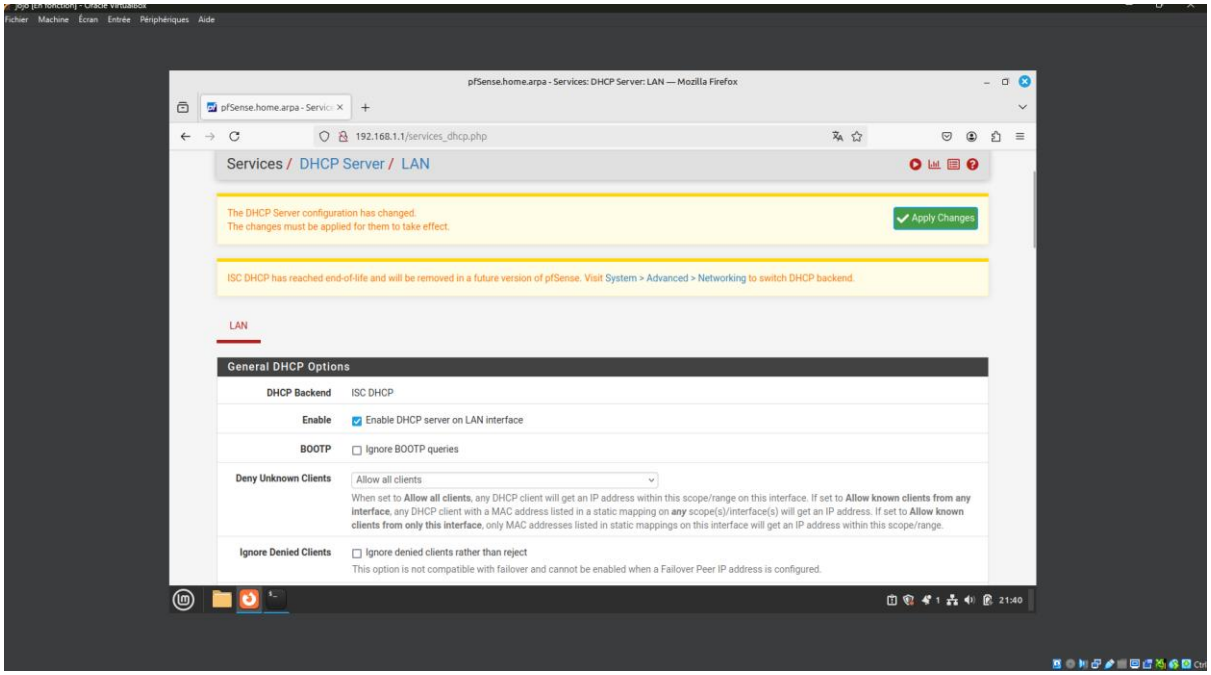


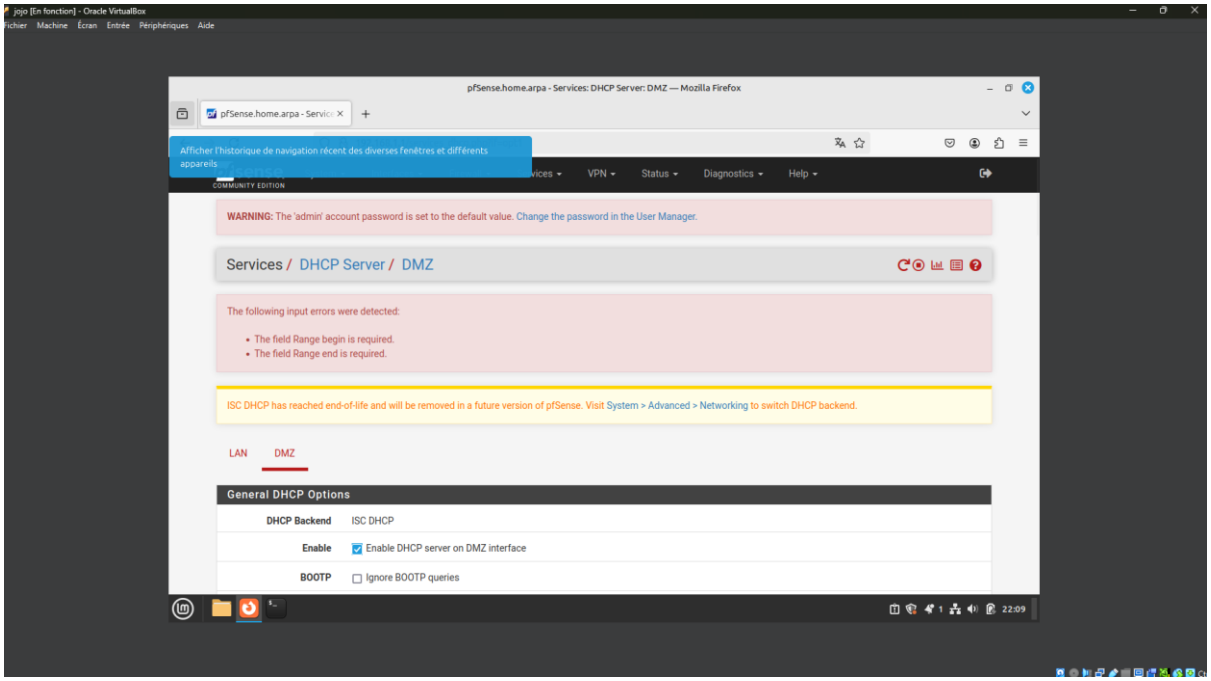
Ensuite nous allons nous connecter a linrerface pour configurer les regles de filtrages





Mtn on va activer le DHCP server sur nos interfaces LAN et DMZ





(j'avais donne l'adresse LAN sur le terminal)

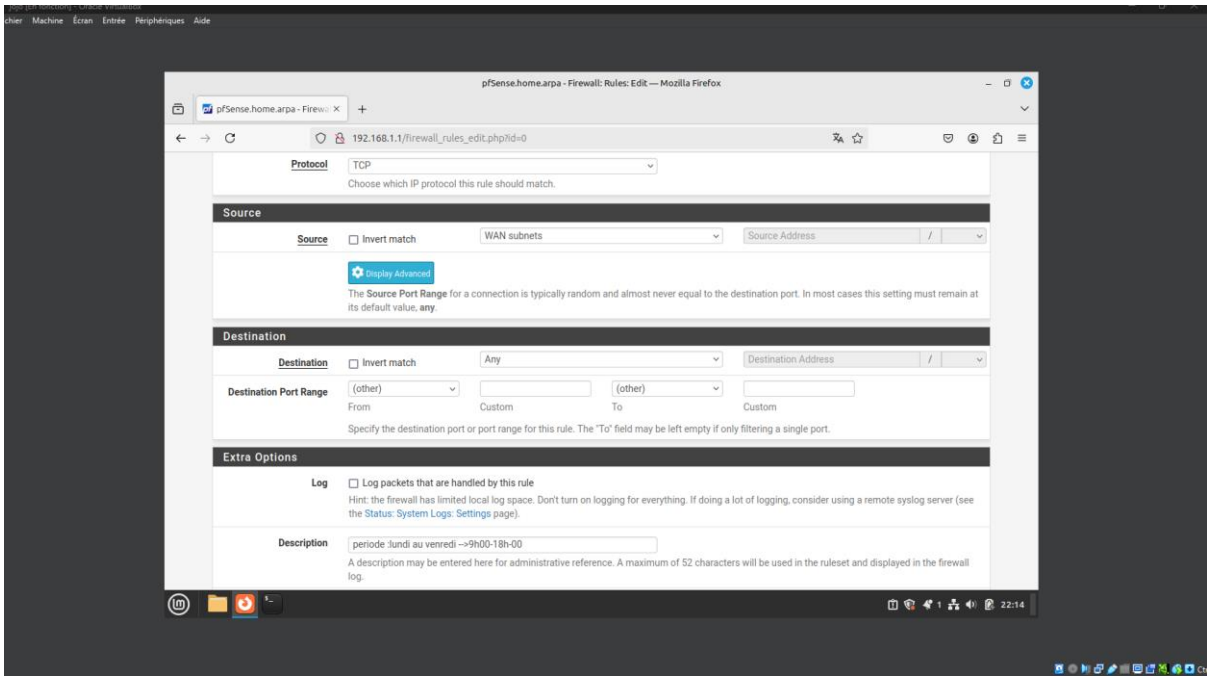
3. Sécurisation du réseau avec des règles de filtrage

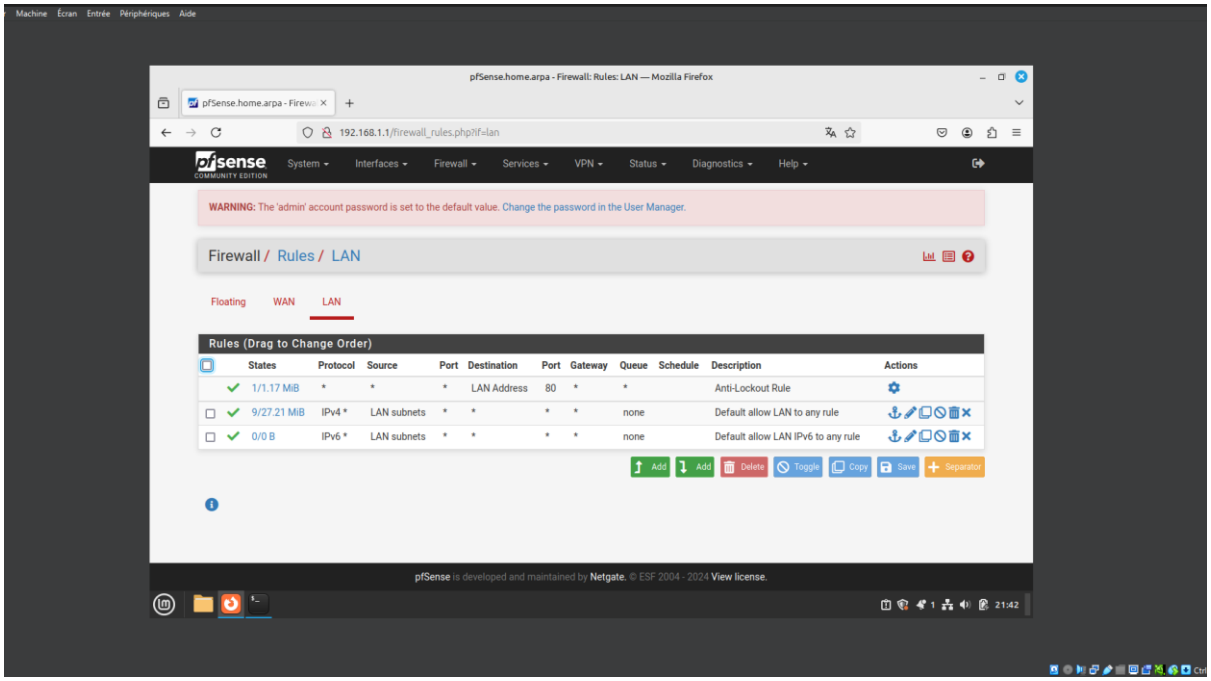
3.1. Création de règles de filtrage pour le LAN

Par défaut, pfSense autorise tout le trafic entrant et sortant sur le réseau LAN

Nous allons bloquer l'accès à Internet pendant les heures de travails

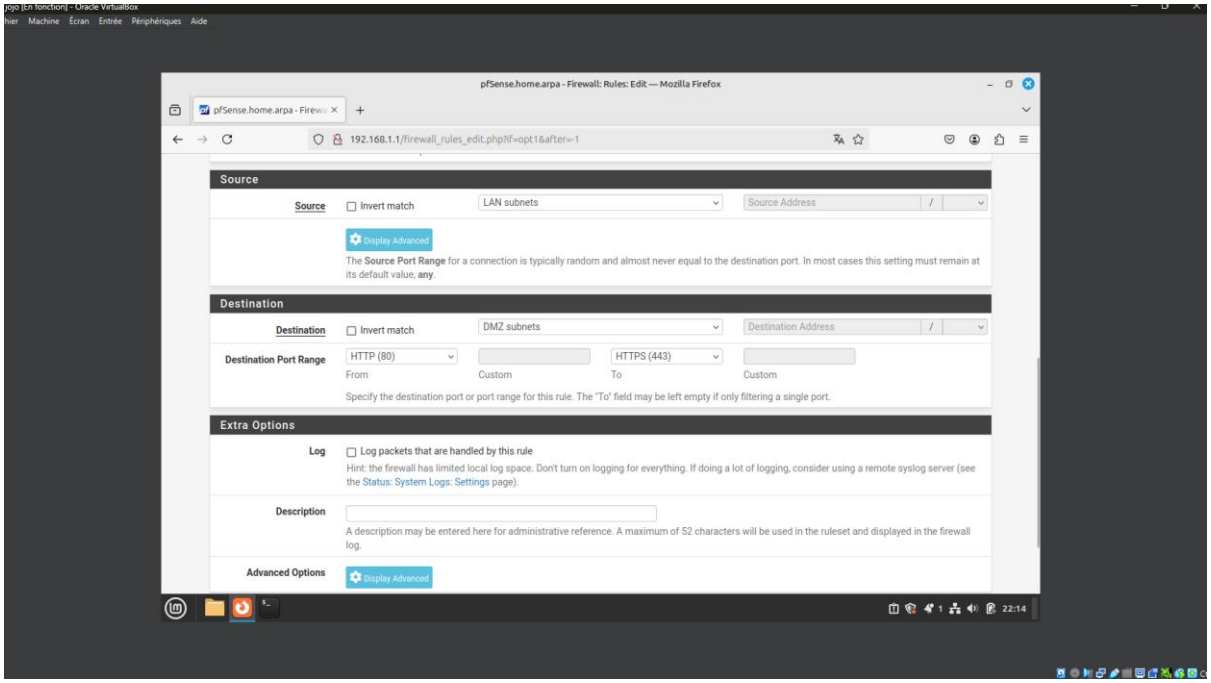
SUR LE LAN

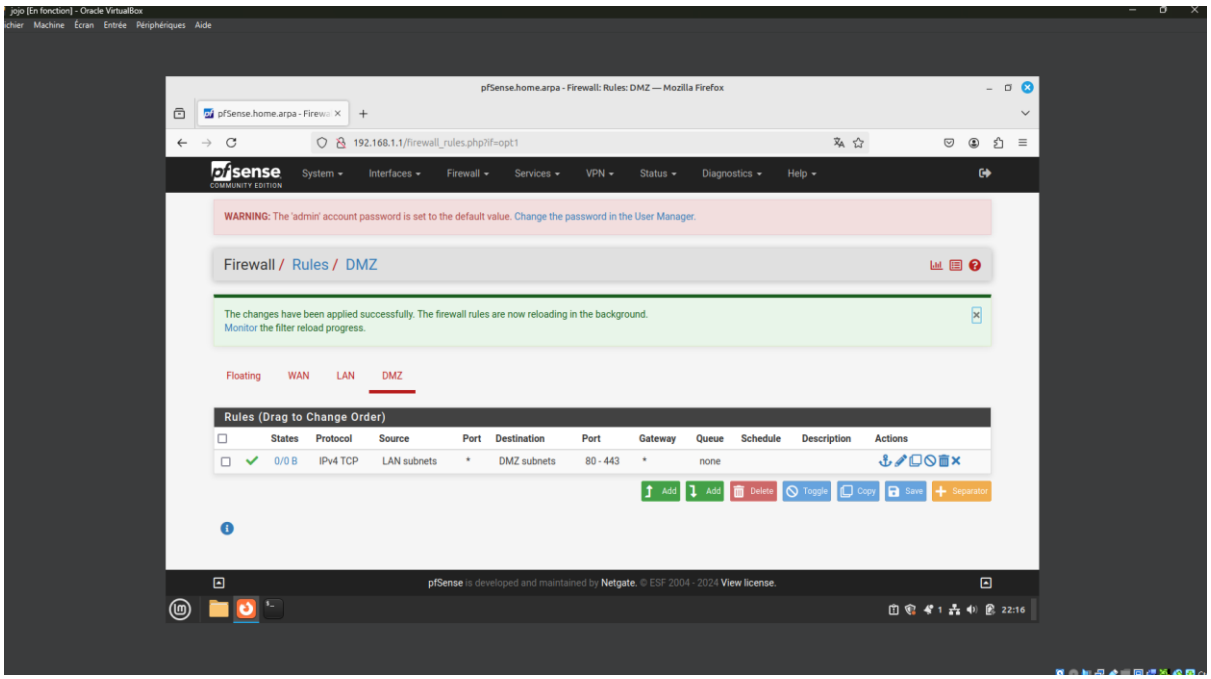




Maintenant on va créer une règle pour autoriser l'accès HTTP (port 80) et HTTPS (port 443) au serveur web dans la DMZ depuis le réseau LAN.

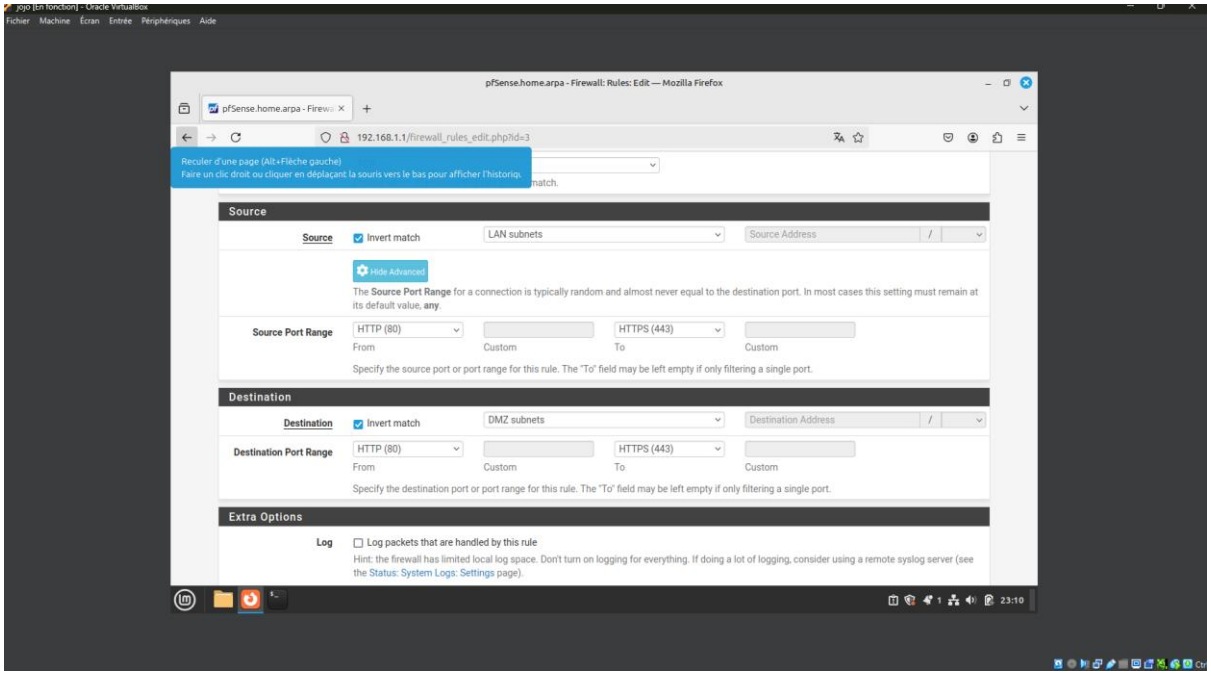
SUR DMZ



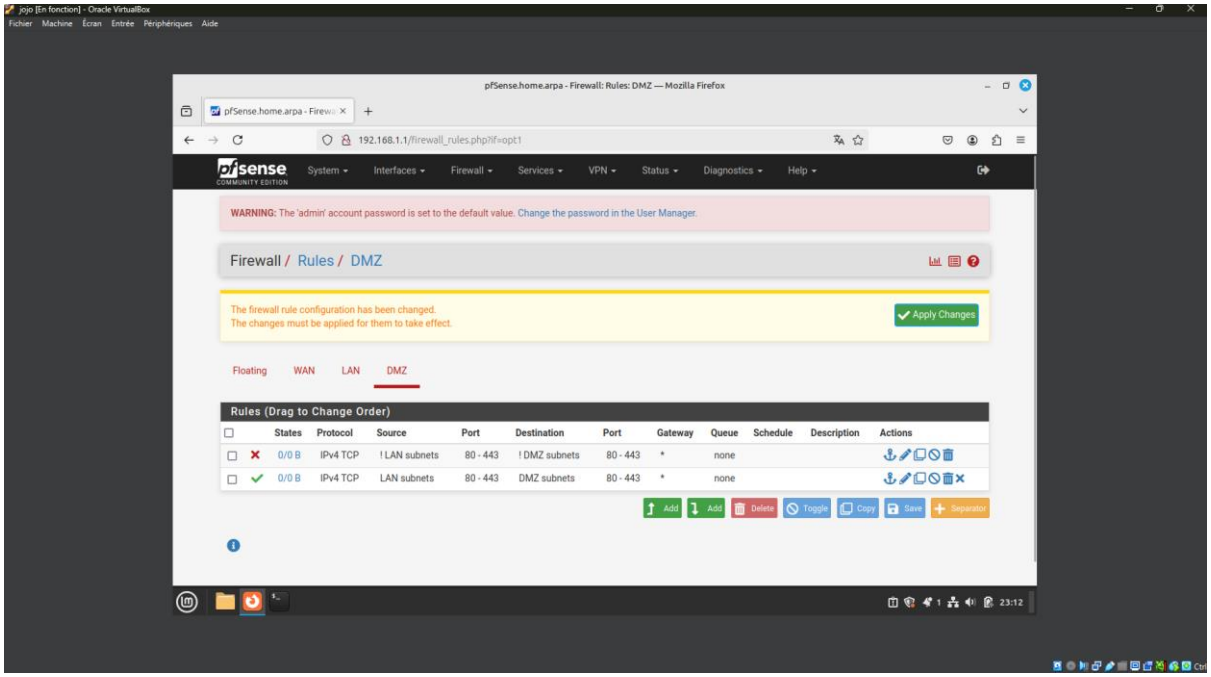


3.2. Filtrage entre le LAN et la DMZ

Maintenant nous allons créez une règle qui autorise uniquement l'accès HTTP/HTTPS depuis le LAN vers le serveur de la DMZ.

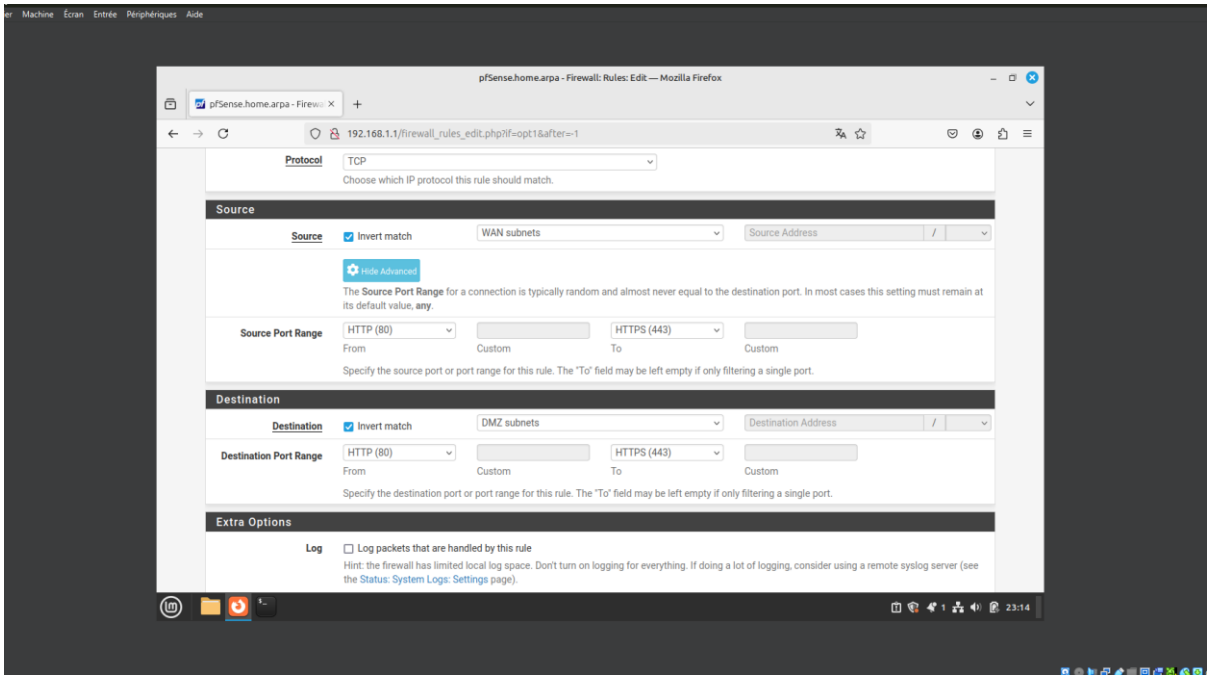


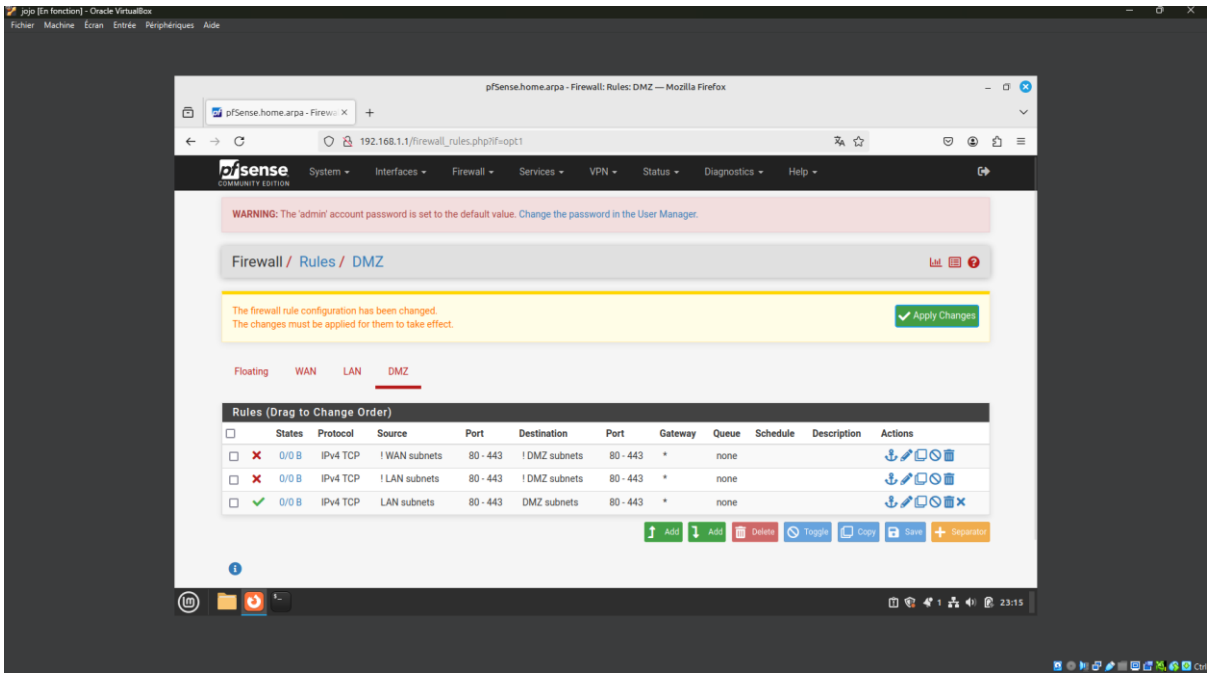
(”Invert ”match sert à bloquer tout autre trafic excepte ceux sélectionne)



3.3. Règles sur la DMZ et le WAN

Nous allons autorisez uniquement le trafic HTTP/HTTPS depuis le WAN vers la DMZ pour permettre l’accès aux services web et Bloquez tout autre trafic entrant sur la DMZ.

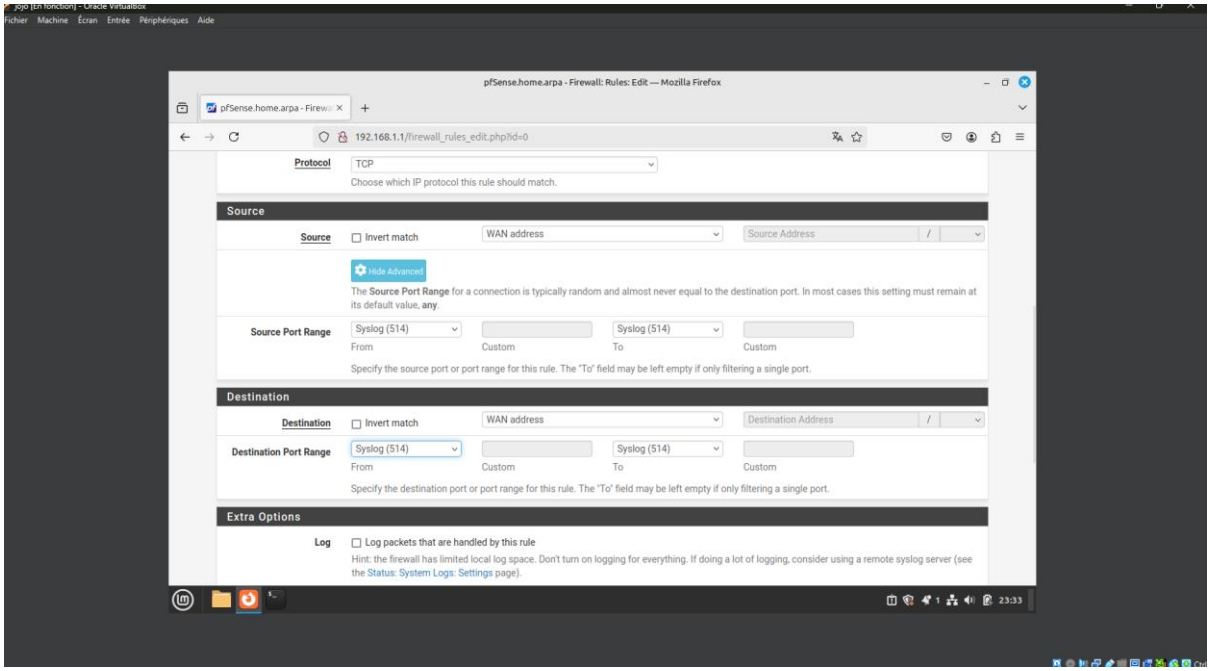


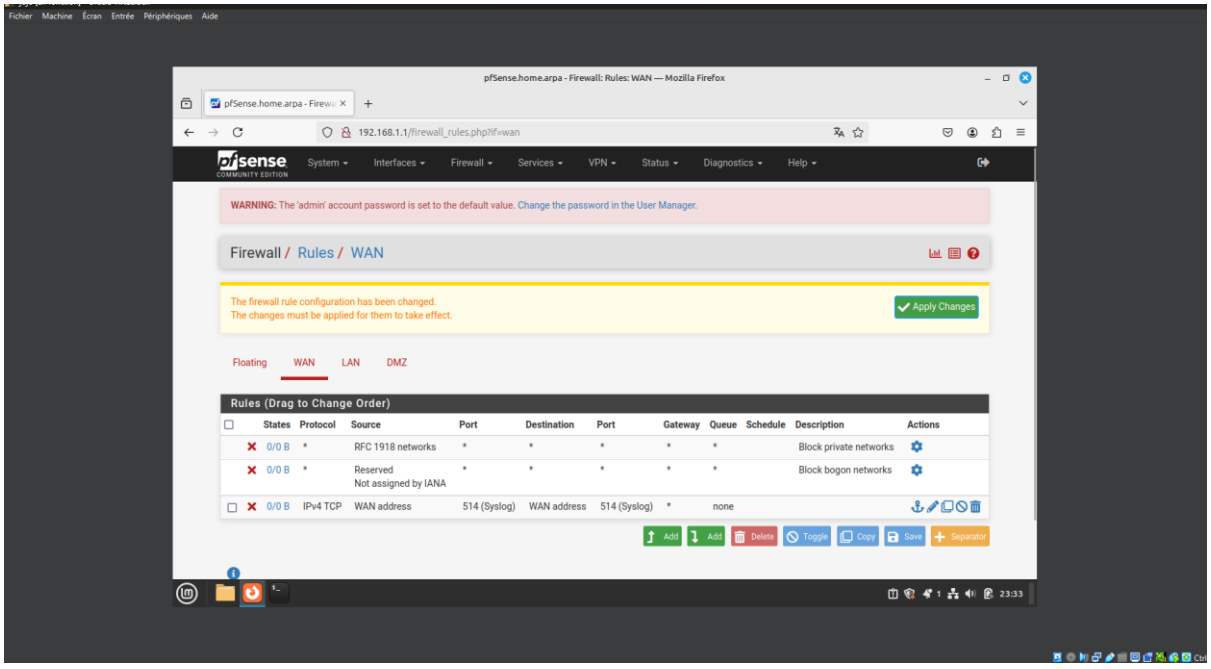


3.4. Règles de filtrage pour le WAN

Nous allons faire en sorte que le pare-feu refuse l'accès Internet en fonction de certains critères (par exemple, bloquer certains sites)

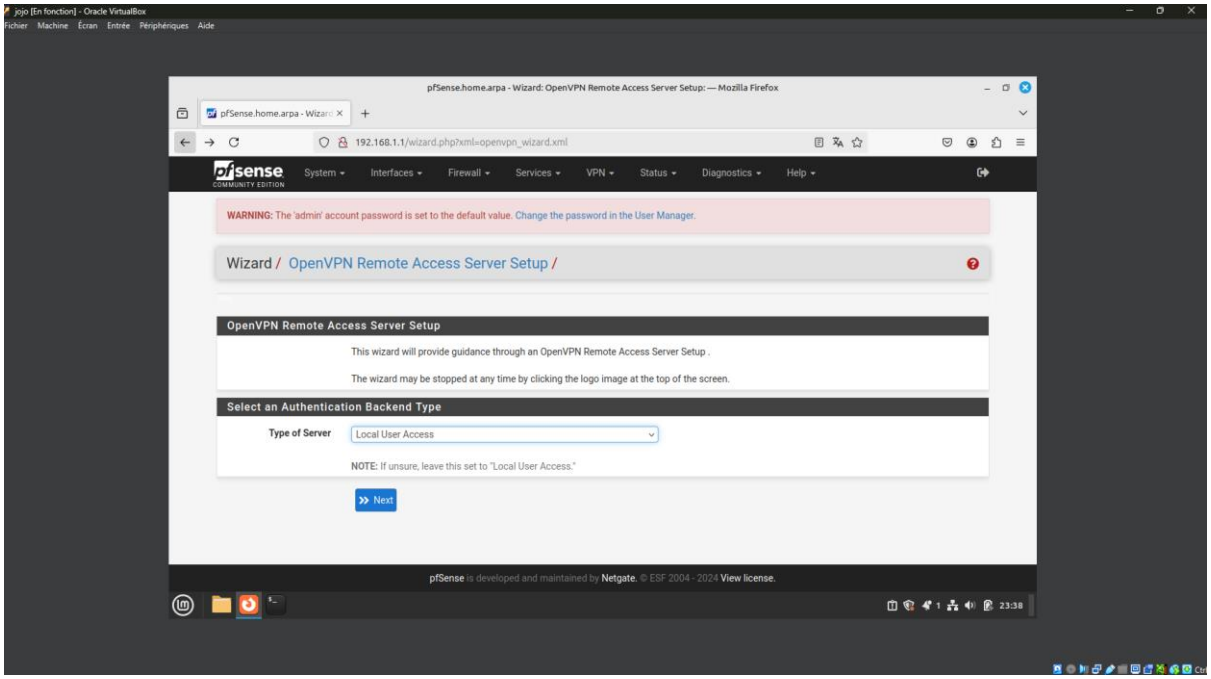
Dans ce cas nous avons utiliser syslog

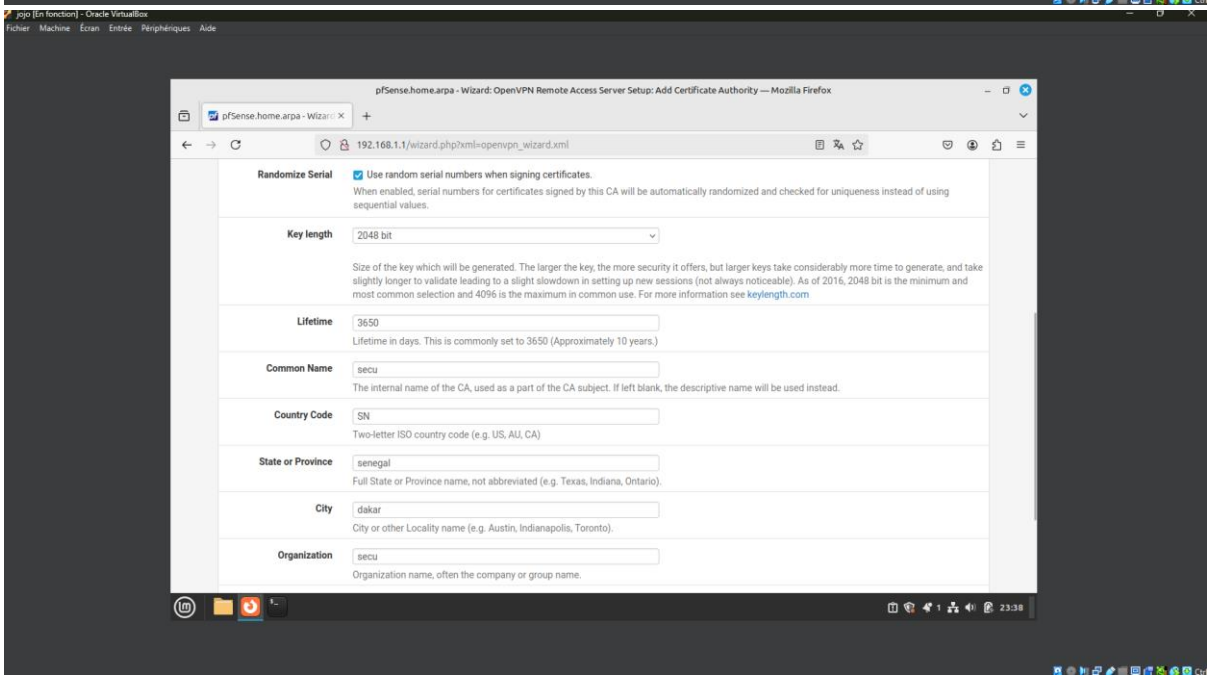


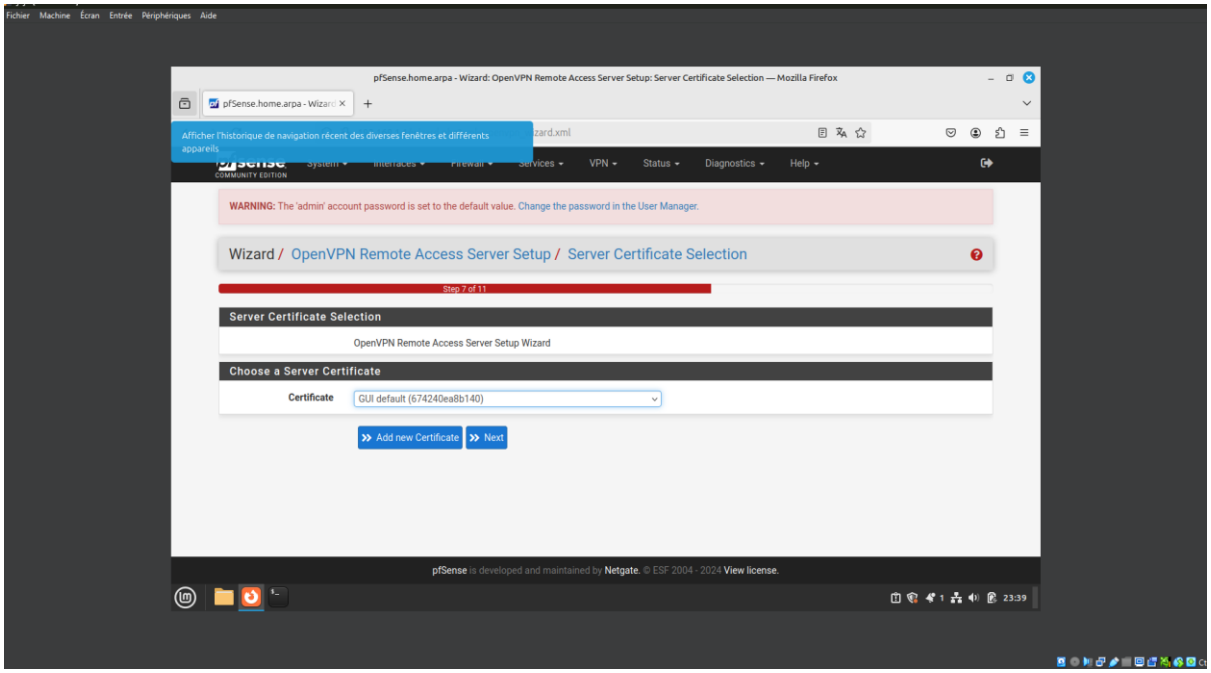


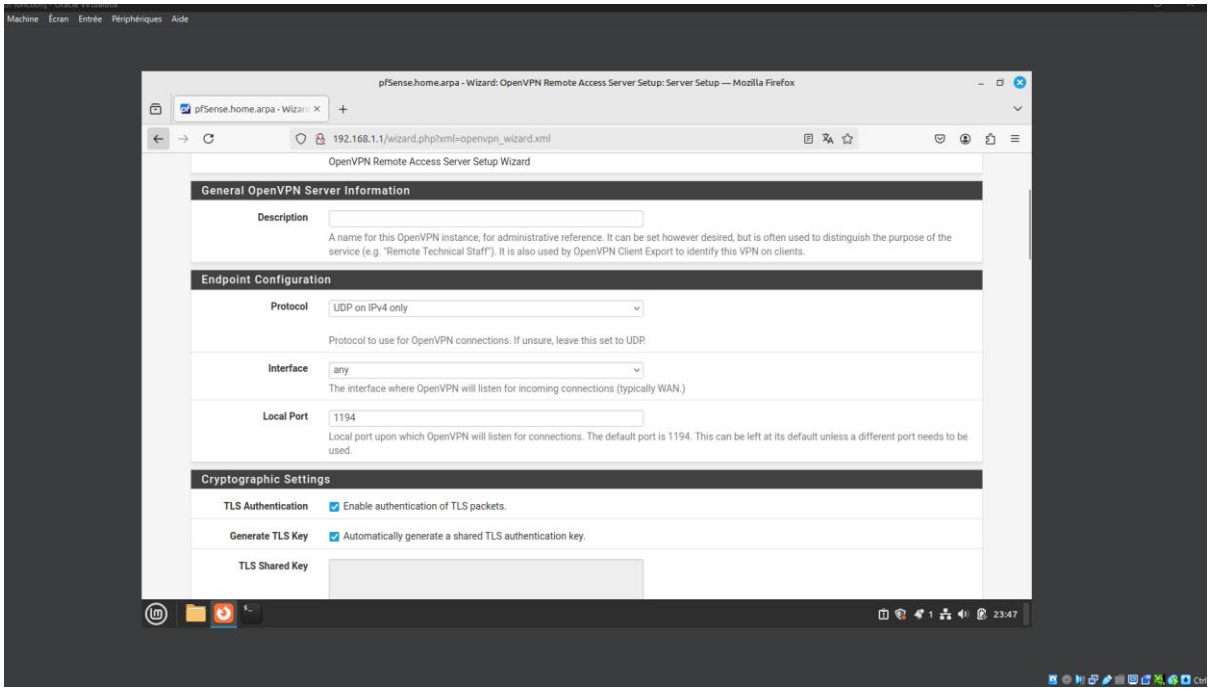
4)Configuration d’un VPN

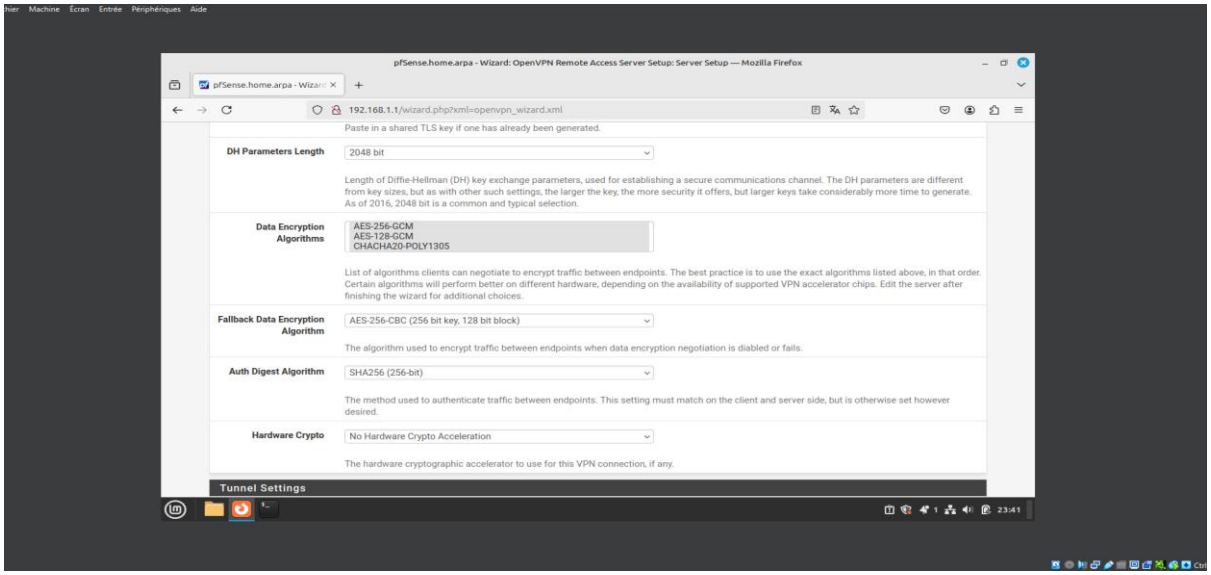
4.1. Configurer un VPN (OpenVPN) pour un accès sécurisé

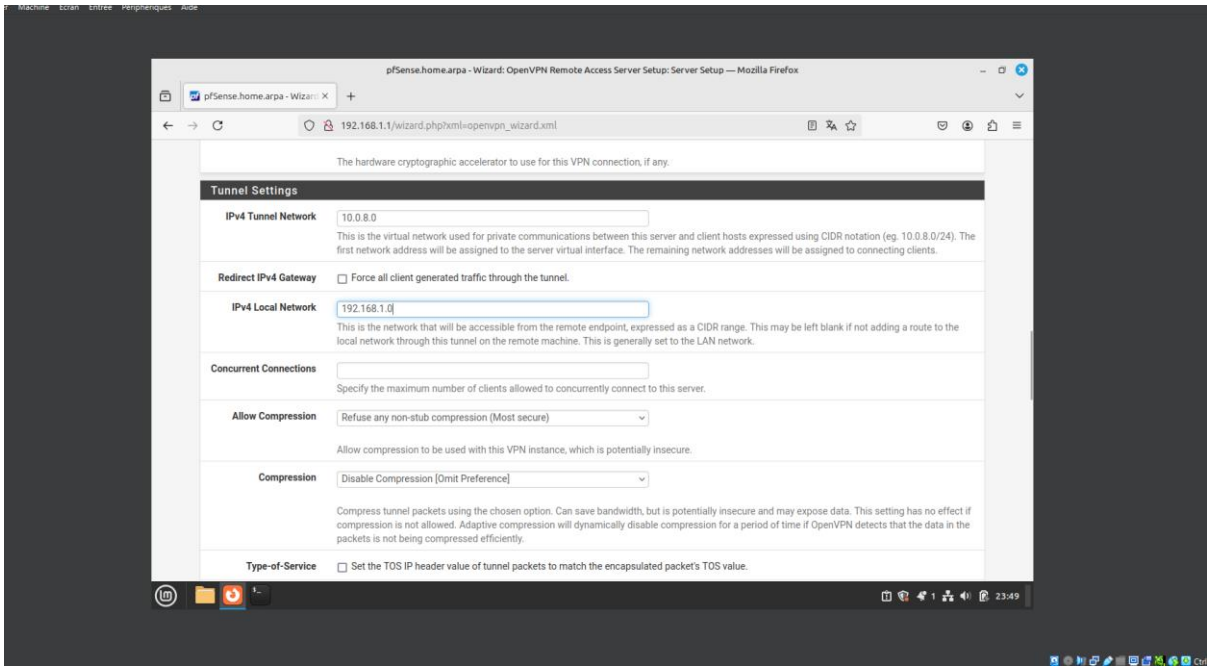

jojo [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
pfSense.home.arpa - Wizard: OpenVPN Remote Access Server Setup: — Mozilla Firefox
pfSense.home.arpa - Wizard
192.168.1.1/wizard.php?xml=openvpn_wizard.xml
pfsense COMMUNITY EDITION
System Interfaces Firewall Services VPN Status Diagnostics Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Wizard / OpenVPN Remote Access Server Setup /
OpenVPN Remote Access Server Setup
This wizard will provide guidance through an OpenVPN Remote Access Server Setup .
The wizard may be stopped at any time by clicking the logo image at the top of the screen.
Select an Authentication Backend Type
Type of Server
Local User Access
NOTE: If unsure, leave this set to "Local User Access."
Next
pfSense is developed and maintained by Netgate. © ESF 2004 - 2024 View license.
23:38

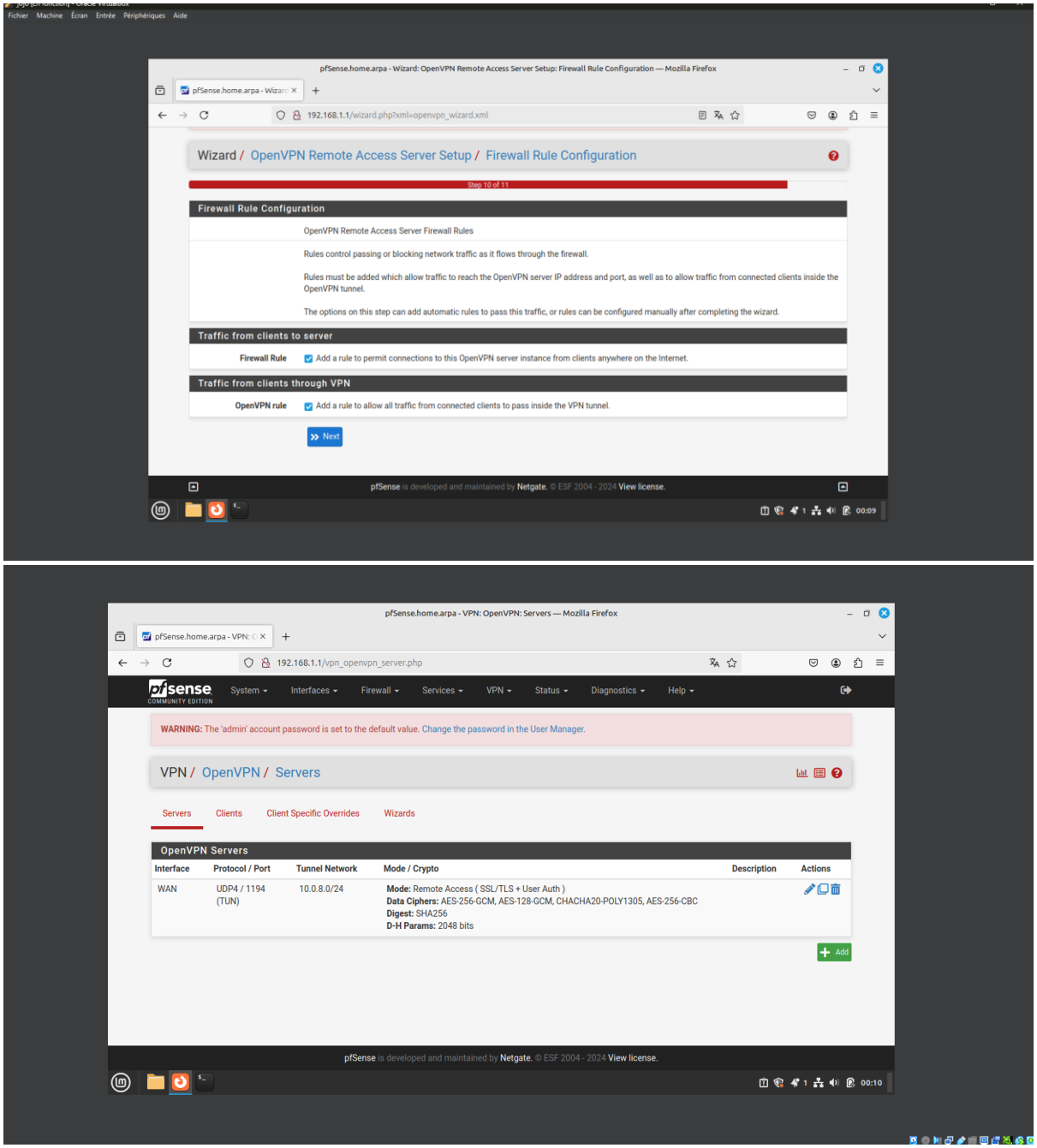

jojo [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
pfSense.home.arpa - Wizard: OpenVPN Remote Access Server Setup: Add Certificate Authority — Mozilla Firefox
pfSense.home.arpa - Wizard
192.168.1.1/wizard.php?xml=openvpn_wizard.xml
Randomize Serial
Use random serial numbers when signing certificates.
When enabled, serial numbers for certificates signed by this CA will be automatically randomized and checked for uniqueness instead of using sequential values.
Key length
2048 bit
Size of the key which will be generated. The larger the key, the more security it offers, but larger keys take considerably more time to generate, and take slightly longer to validate leading to a slight slowdown in setting up new sessions (not always noticeable). As of 2016, 2048 bit is the minimum and most common selection and 4096 is the maximum in common use. For more information see keylength.com
Lifetime
3650
Lifetime in days. This is commonly set to 3650 (Approximately 10 years.)
Common Name
secu
The internal name of the CA, used as a part of the CA subject. If left blank, the descriptive name will be used instead.
Country Code
SN
Two-letter ISO country code (e.g. US, AU, CA)
State or Province
senegal
Full State or Province name, not abbreviated (e.g. Texas, Indiana, Ontario).
City
dakar
City or other Locality name (e.g. Austin, Indianapolis, Toronto).
Organization
secu
Organization name, often the company or group name.
23:38


pfSense.home.arpa - Wizard: OpenVPN Remote Access Server Setup: Server Certificate Selection — Mozilla Firefox
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Wizard / OpenVPN Remote Access Server Setup / Server Certificate Selection
Step 7 of 11
Server Certificate Selection
OpenVPN Remote Access Server Setup Wizard
Choose a Server Certificate
Certificate
GUI default (674240ea8b140)
Add new Certificate
Next
pfSense is developed and maintained by Netgate. © ESF 2004 - 2024 View license.


pfSense.home.arpa - Wizard: OpenVPN Remote Access Server Setup: Server Setup — Mozilla Firefox
192.168.1.1/wizard.php?xml=openvpn_wizard.xml
OpenVPN Remote Access Server Setup Wizard
General OpenVPN Server Information
Description
A name for this OpenVPN instance, for administrative reference. It can be set however desired, but is often used to distinguish the purpose of the service (e.g. "Remote Technical Staff"). It is also used by OpenVPN Client Export to identify this VPN on clients.
Endpoint Configuration
Protocol
UDP on IPv4 only
Protocol to use for OpenVPN connections. If unsure, leave this set to UDP.
Interface
any
The interface where OpenVPN will listen for incoming connections (typically WAN.)
Local Port
1194
Local port upon which OpenVPN will listen for connections. The default port is 1194. This can be left at its default unless a different port needs to be used.
Cryptographic Settings
TLS Authentication
Enable authentication of TLS packets.
Generate TLS Key
Automatically generate a shared TLS authentication key.
TLS Shared Key

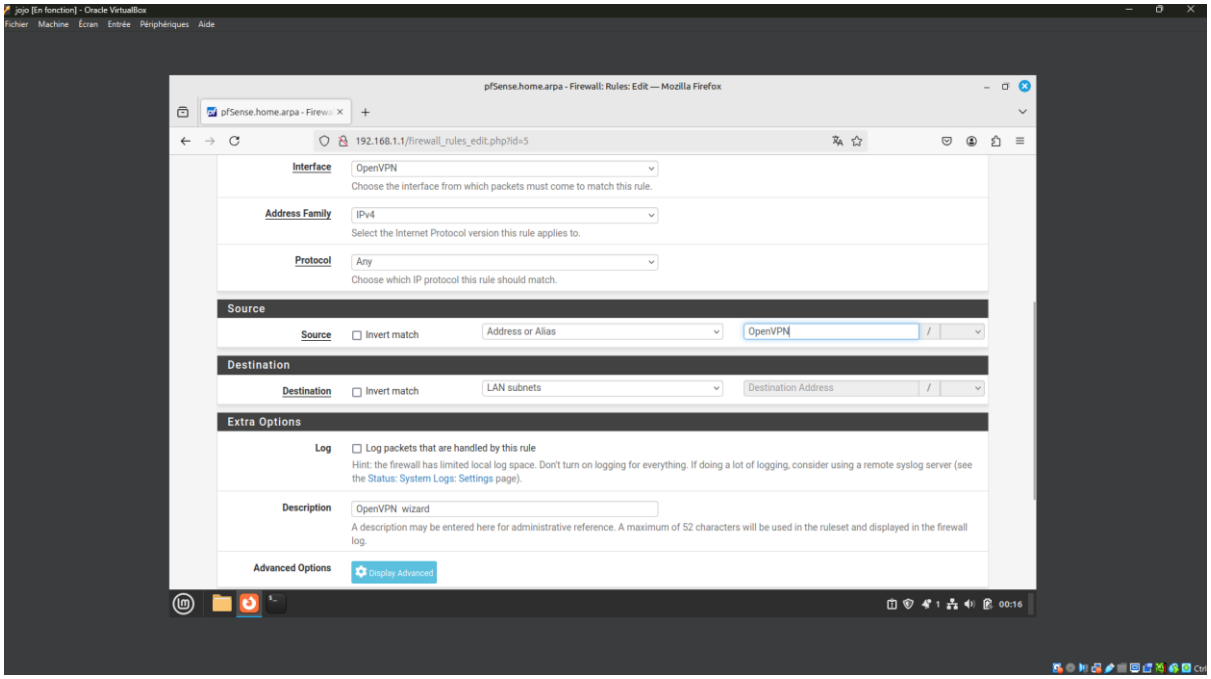

pfSense.home.arpa - Wizard: OpenVPN Remote Access Server Setup: Server Setup — Mozilla Firefox
192.168.1.1/wizard.php?xml=openvpn_wizard.xml
Paste in a shared TLS key if one has already been generated.
DH Parameters Length
2048 bit
Length of Diffie-Hellman (DH) key exchange parameters, used for establishing a secure communications channel. The DH parameters are different from key sizes, but as with other such settings, the larger the key, the more security it offers, but larger keys take considerably more time to generate. As of 2016, 2048 bit is a common and typical selection.
Data Encryption Algorithms
AES-256-GCM
AES-128-GCM
CHACHA20-POLY1305
List of algorithms clients can negotiate to encrypt traffic between endpoints. The best practice is to use the exact algorithms listed above, in that order. Certain algorithms will perform better on different hardware, depending on the availability of supported VPN accelerator chips. Edit the server after finishing the wizard for additional choices.
Fallback Data Encryption Algorithm
AES-256-CBC (256 bit key, 128 bit block)
The algorithm used to encrypt traffic between endpoints when data encryption negotiation is diabled or fails.
Auth Digest Algorithm
SHA256 (256-bit)
The method used to authenticate traffic between endpoints. This setting must match on the client and server side, but is otherwise set however desired.
Hardware Crypto
No Hardware Crypto Acceleration
The hardware cryptographic accelerator to use for this VPN connection, if any.
Tunnel Settings

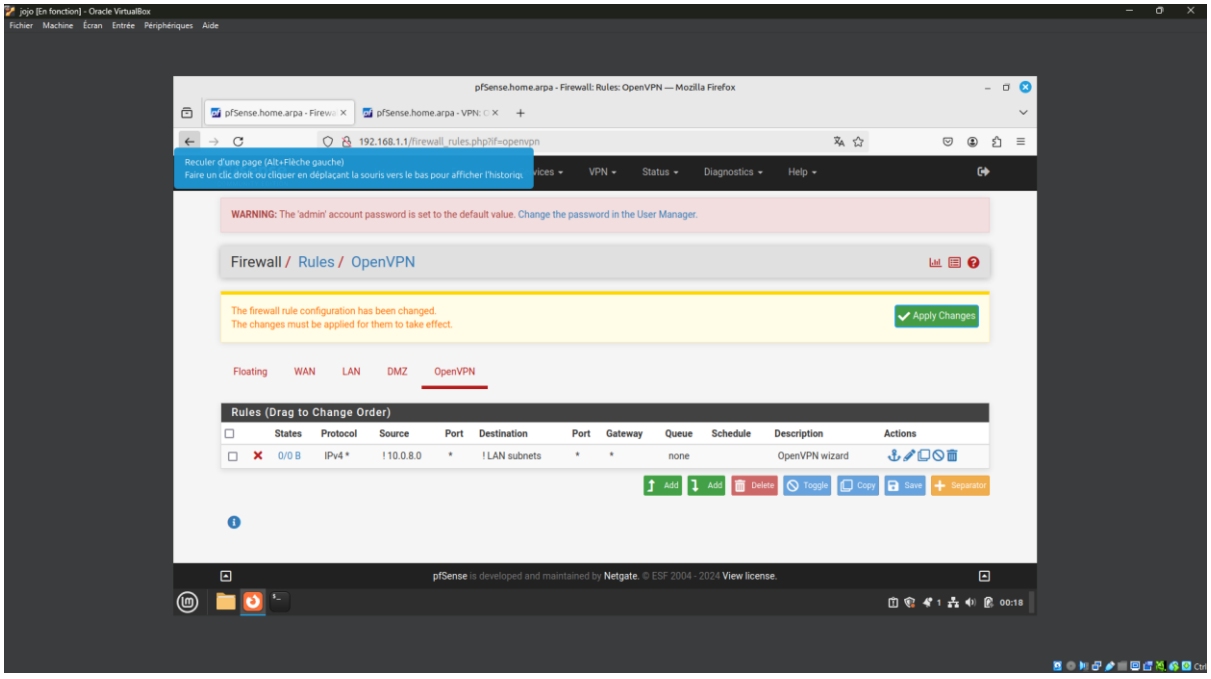

pfSense.home.arpa - Wizard: OpenVPN Remote Access Server Setup: Server Setup — Mozilla Firefox
192.168.1.1/wizard.php?xml=openvpn_wizard.xml
The hardware cryptographic accelerator to use for this VPN connection, if any.
Tunnel Settings
IPv4 Tunnel Network
10.0.8.0
This is the virtual network used for private communications between this server and client hosts expressed using CIDR notation (eg. 10.0.8.0/24). The first network address will be assigned to the server virtual interface. The remaining network addresses will be assigned to connecting clients.
Redirect IPv4 Gateway
Force all client generated traffic through the tunnel.
IPv4 Local Network
192.168.1.0
This is the network that will be accessible from the remote endpoint, expressed as a CIDR range. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.
Concurrent Connections
Specify the maximum number of clients allowed to concurrently connect to this server.
Allow Compression
Refuse any non-stub compression (Most secure)
Allow compression to be used with this VPN instance, which is potentially insecure.
Compression
Disable Compression [Omit Preference]
Compress tunnel packets using the chosen option. Can save bandwidth, but is potentially insecure and may expose data. This setting has no effect if compression is not allowed. Adaptive compression will dynamically disable compression for a period of time if OpenVPN detects that the data in the packets is not being compressed efficiently.
Type-of-Service
Set the TOS IP header value of tunnel packets to match the encapsulated packet's TOS value.



4.2. Création de règles pour le VPN

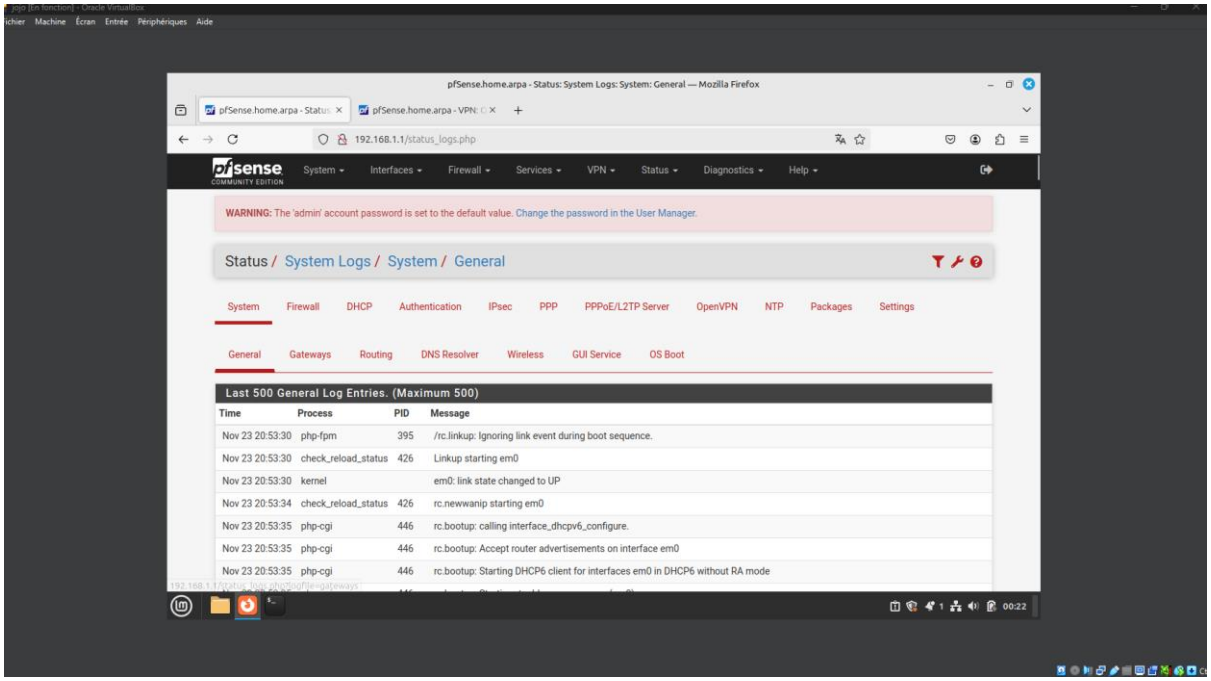
Nous allons creer une regle pour permettre le trafic VPN et autoriser l'accès au réseau interne via le tunnel sécurisé.





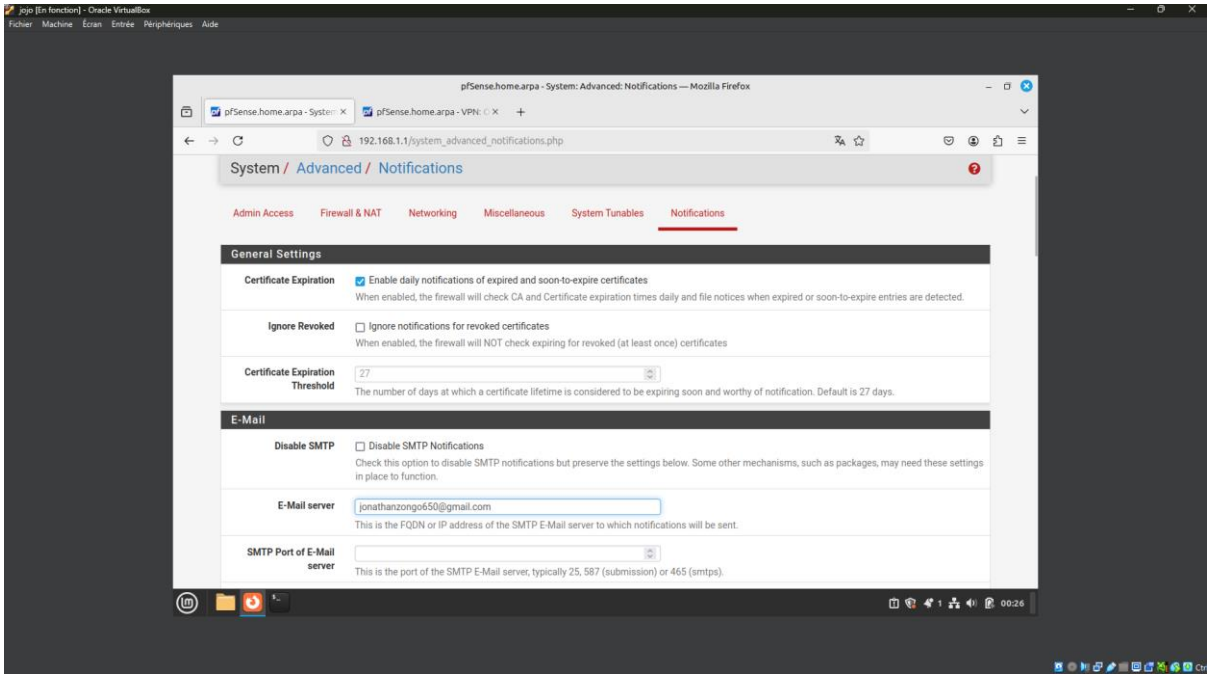
5. Surveillance et rapports

Nous allons aller pour surveiller les logs du pare-feu et vérifier si les règles fonctionnent comme prévu



Maintenant nous allons activez les alertes pour être notifié des tentatives d'accès non autorisées.

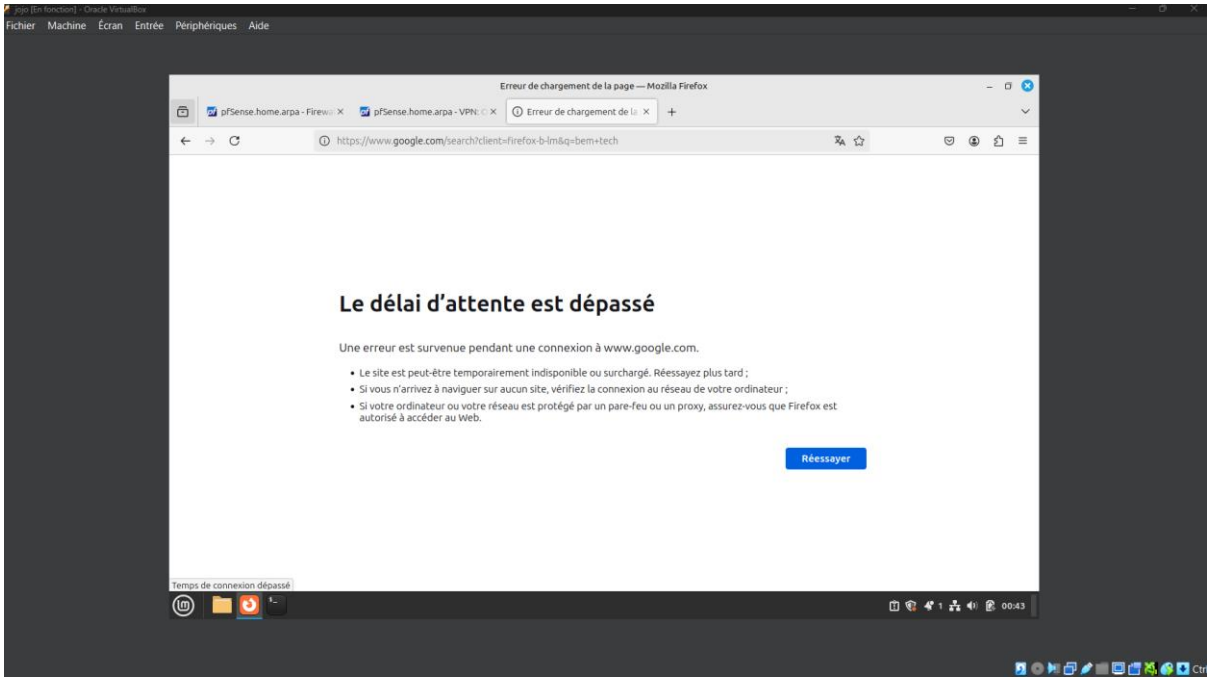
Pour ce faire je vais y ajouter mon adresse mail pour y être notifiée



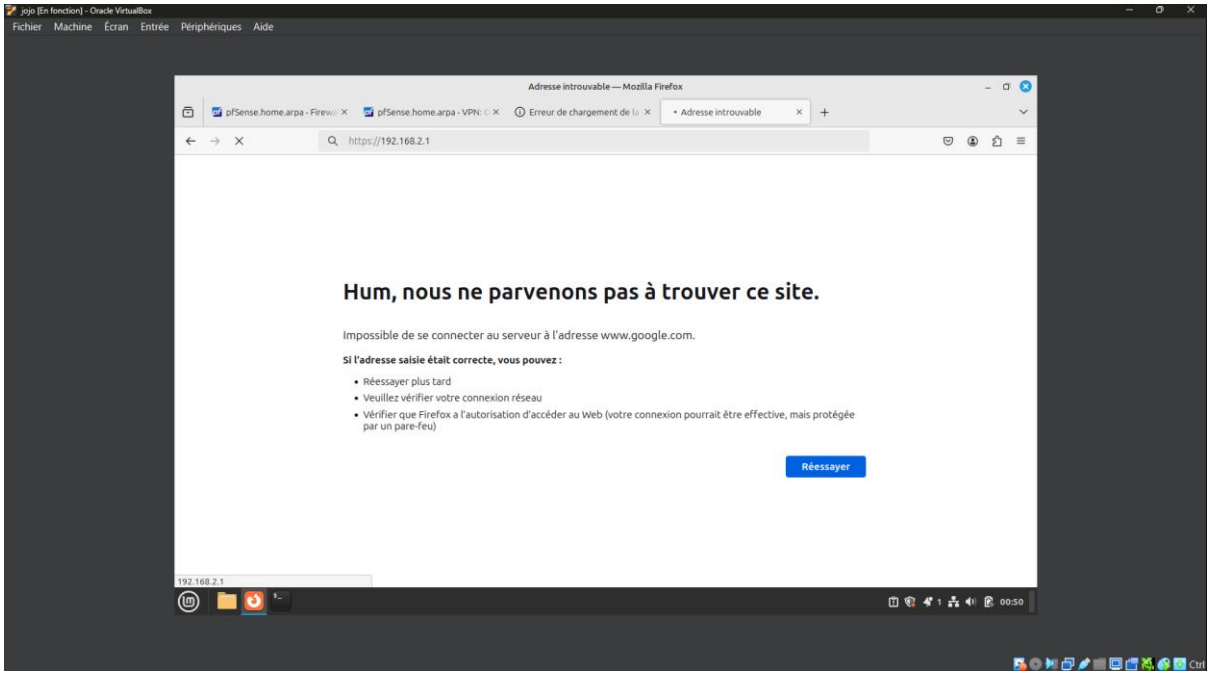
6. Tests de sécurité

Depuis le LAN

Pendant les heures de resrtrictions

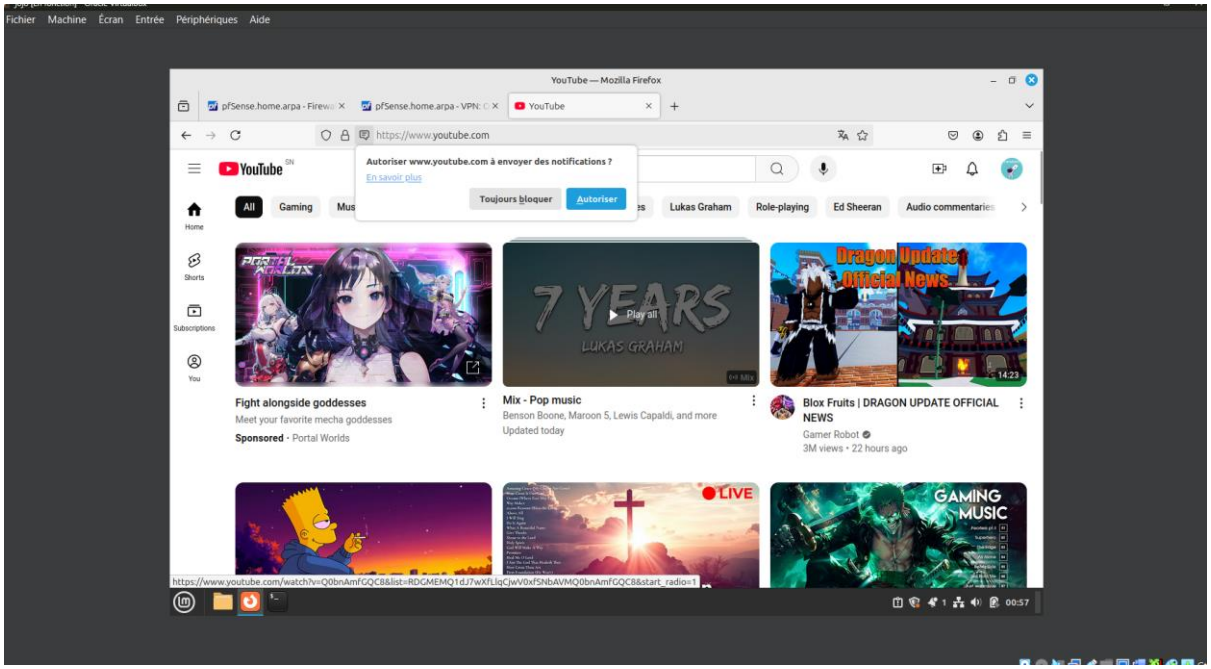


Depuis LAN : `ping 192.168.2.1`



Les règles de filtrage fonctionnent correctement : Par exemple, un trafic bloqué entre LAN et DMZ ne doit pas passer

Depuis DMZ (LAN vers DMZ)



Les services sont accessibles comme prévu : Par exemple, on est autorisés peuvent accéder au serveur web dans la DMZ

(une règle avait été ajoutée)

AVEC VPN